

Human-AI Interactive and Continuous Sensemaking: A Case Study of Image Classification using Scribble Attention Maps

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ABSTRACT

Advances in Artificial Intelligence (AI), especially the stunning achievements of Deep Learning (DL) in recent years, have shown AI/DL models possess remarkable understanding towards the logic reasoning behind the solved tasks. However, human understanding towards what knowledge is captured by deep neural networks is still elementary and this has a detrimental effect on human's trust in the decisions made by AI systems. Explainable AI (XAI) is a hot topic in both AI and HCI communities in order to open up the blackbox to elucidate the reasoning processes of AI algorithms in such a way that makes sense to humans. However, XAI is only half of human-AI interaction and research on the other half - human's feedback on AI explanations together with AI making sense of the feedback - is generally lacking. Human cognition is also a blackbox to AI and effective human-AI interaction requires unveiling both blackboxes to each other for mutual sensemaking. The main contribution of this paper is a conceptual framework for supporting effective human-AI interaction, referred to as interactive and continuous sensemaking (HAICS). We further implement this framework in an image classification application using deep Convolutional Neural Network (CNN) classifiers as a browser-based tool that displays network attention maps to the human for explainability and collects human's feedback in the form of scribble annotations overlaid onto the maps. Experimental results using a real-world dataset

has shown significant improvement of classification accuracy (the AI performance) with the HAICS framework.

CCS CONCEPTS

• **Human-centered computing** → **Collaborative interaction; HCI theory, concepts and models; Heat maps.**

KEYWORDS

interactive sensemaking, explainable AI, image classification, attention map, scribble interaction

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1 INTRODUCTION

Today, humans are using various AI based systems in their everyday life. Routine tasks such as searching the Internet or shopping online all employ some sort of AI or DL techniques. These techniques can make superior judgments or predictions compared to humans on a variety of tasks and a widely cited example is AlphaGo [27], which demonstrates remarkable understanding towards the complicated logic reasoning behind. However, despite algorithmic advances in AI/DL systems, they have just started to interact with humans [2, 6, 13, 21, 25]. A major obstacle to effective human-AI interaction is the blackbox nature of DL models, which typically supply useful results without providing much rationale for their output in such a way that makes sense to humans [34]. This problem led to explainable AI (XAI), a hot topic in the AI community on

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opening up the blackbox to elucidate the reasoning mechanisms of AI/DL algorithms to humans [34]. XAI is also gaining attention in the HCI community and the research has been focused on the need of XAI for decision making and mitigating automation bias [4], the need for AI explanations to be aligned with human mental models [17], visualization and sensemaking of explanations [33], and XAI evaluations [9, 23].

However, despite great achievements in XAI research, there is a critical gap - *human's feedback on AI explanations together with AI making sense of the feedback for continuous improvement* is generally lacking. AI providing explanations to human is only half of human-AI interaction. Human cognition is also a blackbox to AI and effective human-AI interaction requires unveiling both blackboxes to each other for mutual sensemaking. In particular, the human has to be able to feedback to the AI: a) do the explanations make sense to the human? b) is the reasoning of results correct from the human's perspective? and c) how can the AI learn from the human's feedback for continuous improvement? The main contribution of this paper is a conceptual framework for supporting effective human-AI interaction, referred to as interactive and continuous sensemaking (HAICS) and its implementation in an image classification application using deep CNNs. We develop a browser-based visualization tool that allows a human to view network attention maps (CNN explanation) and provide scribble annotations overlaid onto the maps (human feedback). This framework is timely as interactive and continuous sensemaking and its natural interfaces are identified as a major research agenda in a sensemaking research roadmap [30]. Furthermore, the image classification case study with a real-world dataset has clearly shown significant improvement of classification accuracy (the AI performance) with the HAICS framework.

2 HAICS: HUMAN-AI INTERACTIVE AND CONTINUOUS SENSEMAKING FRAMEWORK

This section details the HAICS framework. As illustrated by Fig. 1a, the framework consists of the two blackboxes of AI and human cognition and the three iterative processes of: (1) human sensemaking of AI through AI's explanation, (2) AI sensemaking of human's feedback on its explanation through scribble annotation, and (3) AI fine-tuning its model based on human's annotation feedback. The three iterative processes form a continuous interactive sensemaking cycle. It is easier to elaborate the framework using a case study in which the framework is applied to solving a real world problem. Fig. 1b shows such an example implementation of ship image classification using deep CNN classifiers.

2.1 Human sensemaking of CNN classifiers through network attention maps

A common criticism of CNNs is their opaque learning and reasoning process. For image based learning tasks, the explainability of a learning model can be measured by its ability to accurately localize the interested image objects. To uncover the captured knowledge, one way is to visualize the "attention" of CNNs [26, 28, 29, 35–37] as heatmaps, which is also commonly used to demonstrate that a network has appropriately (aligning with human gaze) localized the target object or patterns. Class Activation Mapping (CAM) [37]

and Gradient-weighted CAM (Grad-CAM) [26] do not explicitly add attention layers, which can work off-the-shelf with the state-of-the-art CNN classifiers such as ResNet [12]. In addition, it has been reported that CAM based attention maps can serve as localization cues for weakly supervised semantic segmentation [15].

The attention visualization methods provide a great aid to the transparency of CNNs and can unveil their internal reasoning mechanism to the humans in a way that makes sense to them. For example, Fig. 1b (1) shows a correctly classified military vessel with a heatmap overlay that reflects what the model is focusing on with redder regions denoting more focused areas and bluer less focused (analogous to how human recognizes objects). It also appears that the attention on the central part of the ship provides a correct explanation because the redder region indeed contains more discriminative features of a military vessel such as high mast, unique shape and layout on board. On the other hand, the bluer areas mostly cover the image backgrounds such as sea and sky.

2.2 AI sensemaking of human feedback through scribble annotations overlaid onto attention maps

How does the AI know whether the human agrees that the CNN classifier has accurately localized the interested objects in an attention map and therefore trust its classification result? What if the human does not agree? If the human finds a visualized network attention not properly localized, it is important for them to feedback this information to the classifier so that the network will learn from it and do better in the subsequent iterations. Human attention supervised networks are able to achieve better performance for the following two reasons: a) multiple contextual patterns and object parts could be discriminative to a learning task and any of these can be focused on by the network due to randomness of network training; and b) humans possess a vast amount of professional expertise or life experience, which helps identify the most discriminative image region that is more generalizable to the unseen test set.

Scribble labelling is widely used in weakly annotated semantic segmentation tasks, with a stroke indicating the foreground object and another indicating the background, which effectively reduce the annotation cost. There have been interactive segmentation methods utilizing graph cuts based algorithms [5, 10, 11, 19, 24] to find the appropriate object boundary. However, recently scribble type annotations have been used as weak training labels for deep learning segmentation models [20, 32]. In particular, Tang et al. [32] proposed using Conditional Random Field (CRF) as a loss function to determine the object boundary during network training.

In this work, we explore the effectiveness and practicability of regularizing the attention of CNN classifiers with human provided feedback by means of scribble annotations overlaid onto the attention maps. Human is allowed to make two types of annotation scribbles: 1) a positive attention mark (in red) indicating the image region that the network attention should be at, and 2) a negative attention mark (in green) indicating the image region that should be ignored by the network. The scribbles are then recorded for fine-tuning of the inspected network. For example, Fig. 1b (2) displays a 'Carrier' ship that was wrongly classified by the network with a

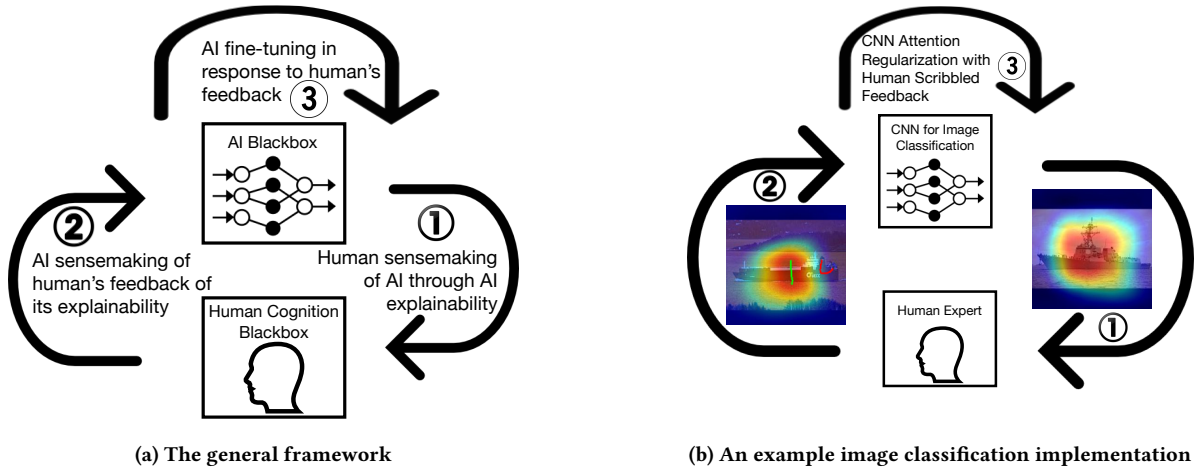


Figure 1: The Human-AI Interactive and Continuous Sensemaking (HAICS) Framework



Figure 2: The browser interface of the visualization tool for collecting user scribble attention annotations.

predicted class of ‘Cargo’. Intuitively, the attention on the middle part of the vessel seems to be wrong as it does not contain discriminative features. The human can annotate the incorrect attention map (the explanation) using scribbles (green as ‘remove attention’ and red as ‘add attention’) and then feedback to the CNN classifier for fine-tuning.

We have developed a browser-based network attention visualization tool, which allows the user to view visualized CAM heatmaps (CNN explanation) and provide scribble annotations/corrections (human feedback). The CAM heatmaps guide the user to directly judge on the potentially “misplaced” and “missed” attentions by

the machine, which effectively reduces the cost of labelling. As shown in Fig. 2, the interface allows the user to upload images or a CSV file containing image file paths and then displays one image at a time. The selected image by the user is then sent to a Pytorch [22] back-end running a user supplied network model in evaluation/testing mode. The back-end processes the image and then returns the predicted class label, top- x class probabilities, and the respective CAM heatmaps (bicubic interpolated) back to the front-end, where x represents a user adjustable parameter with the flexibility of reducing the system response time for large multi-label classification tasks such as the ImageNet [7].

The CAM heatmaps are overlaid onto the input image and displayed to the user in the front-end. The overlay transparency is controlled by an opacity slider in the browser front-end to adjust the alpha value of the heatmap. Finally, the user may choose to draw on any of the x supplied classwise heatmaps (displayed in the ‘Scribble Annotation Window’) by clicking on the displayed classes at the bottom right of the browser page. After clicking the “→” submit button, the coordinates of the drawn scribbles will be sent back to the Pytorch backend to be reconstructed as scribble maps stored on the disk.

2.3 AI fine-tuning CNN models through scribble attention regularization

To avoid dense pixel labelling and annotating a large amount of images, the tool provides the user effective ways of inspecting the network attention and making relevant attention annotations. In particular, to experiment with the tool, an image dataset is split into three parts. The training set would be used to train an initial model offline for inspection, and the validation set would be used online with the interface for collecting the attention scribbles drawn on those images. To determine the order of online annotations, images are sorted in ascending order of the predicted probability of its ground truth class. Hence, the incorrectly classified images and the images with low confidence can be annotated first to save the searching effort. Furthermore, we propose a guideline/strategy for the user to only annotate the ground truth class, with both positive and negative attention scribbles. During the fine-tuning phase, the same scribble mask is replicated to supervise all classes. The labelling effort can be greatly saved while a more focused attention can be generated across classes in this way.

Fig. 1b (3) displays the core mechanism of AI fine-tuning elaborated as follows. Global Average Pooling (GAP) is widely used in the state-of-the-art deep learning classification models such as the GoogLeNets [31] and ResNets [12]. In a nutshell, GAP collapses the spatial dimensions of the last convolution activation maps, which generates a channel-width long image encoding for the softmax classification layer.

Assuming that the last convolutional feature map $\mathbf{F}$ has the size of $H \times W \times K$ and $\mathbf{F}_{ijk}$ denotes an element in $\mathbf{F}$ with respect to the subscripted indices, the GAP operation is

$$\mathbf{f} = \frac{1}{H \times W} \sum_{i \in H} \sum_{j \in W} \mathbf{F}_{ij},$$

where $\mathbf{f} \in \mathbb{R}^K$ is the input feature vector for the softmax classification layer. Let us also denote the number of classes as C , the weight matrix of the classification layer as $\mathbf{W} \in \mathbb{R}^{K \times C}$ and bias as $\mathbf{b} \in \mathbb{R}^C$, the computation of the class probability distribution $\mathbf{p} \in \mathbb{R}^C$ is carried out as:

$$\mathbf{p} = \text{softmax}(\mathbf{W}^T \mathbf{f} + \mathbf{b}).$$

For each class c , the corresponding CAM $\mathbf{M}_c$ is computed by removing the average pooling operation and directly weighting $\mathbf{F}_k \in \mathbb{R}^{H \times W}$ by the corresponding weight elements in $\mathbf{W}_c$:

$$\mathbf{M}_c = \sum_{k \in K} \mathbf{W}_{kc} \cdot \mathbf{F}_k.$$

Under the CNN evaluation mode, $\mathbf{M}_c$ can be up-sampled to the input size, normalized between zero and one by $\min(\mathbf{M}_c)$ and $\max(\mathbf{M}_c)$, and then overlaid to the input image as a heatmap to indicate the most probable location of c .

Let an obtained scribble attention mask $\mathbf{S}_c$ contain the positive attentions marked as 1s and the negative attentions as 0s. Besides, the unannotated pixels on $\mathbf{S}_c$ are marked as a number $e \notin \{0, 1\}$. To use $\mathbf{S}_c$ as a supervision target, during the CNN training mode, $\mathbf{M}_c$ is first up-sampled to the input image size using the bicubic interpolation function g and then applied the sigmoid function over the class dimension for producing coarse pixel-wise activation map comparable to the binary $\mathbf{S}_c$:

$$\mathbf{M}'_c = \text{sigmoid}(g(\mathbf{M}_c + \mathbf{b}_c)),$$

where the bias term is also added to be consistent with the original classification.

Therefore, the new supervising loss function ℓ for the network becomes a composition of two terms: the (left) original classification cross-entropy loss and the (right) *attention regularization loss* in the form of binary cross entropy:

$$\ell = \mathcal{H}(\mathbf{p}, \mathbf{q}) + \frac{\lambda}{N} \sum_{i,j,c} \mathbb{1}(\mathbf{S}_{ijc} \neq e) \mathcal{H}(\mathbf{M}'_{ijc}, \mathbf{S}_{ijc}),$$

where $\mathcal{H}(\cdot, \cdot)$ denotes the cross entropy function, λ denotes a weighting parameter to control the influence of the attention regularization loss, $\mathbf{q}$ denotes the classification one-hot label vector, $\mathbb{1}(\cdot)$ denotes the indicator function and $N = \sum_{i,j,c} \mathbb{1}(\mathbf{S}_{ijc} \neq e)$ represents the normalizer that excludes the unannotated pixels.

3 EXPERIMENTAL VALIDATION

We validate that, with HAICS, AI’s performance is greatly improved after fine-tuning the CNNs to incorporate human’s feedback. In particular, we demonstrate that by making sense of human’s scribble attention maps, classification accuracy of the CNNs is statistically significantly improved using a real-world scaphoid fracture dataset [18], while with an alternative bounding box annotation style, the improvement is not statistically significant.

The scaphoid is one of the eight carpal bones of the wrist. Fractures of the scaphoid are the most common carpal bone injuries [3]. Radiography imaging is a routine examination for detecting the scaphoid fracture, however not all scaphoid fractures can be detected from radiographs, as some cases may require further MR or CT exams for confirmation. We first investigate a challenging scaphoid fracture dataset prepared by Langerhuizen et al. [18], which contains 300 patients, 150 having scaphoid fractures and the rest half having no fracture.

Each patient has a number of 2 to 4 anonymized radiograph images including the posteroanterior (PA), PA ulnar deviation, oblique, and lateral views, resulting in a total of 1197 images. We randomly split the training, validation, and test sets in a 6:2:2 ratio, resulting in 180, 60, and 60 patients respectively and subsequently 719, 238, and 240 images for each set. Each image of a patient inherits the fracture label of the patient, but there is no guarantee that the fracture is visible from an individual view. Furthermore, out of the 150 fracture patients, 127 cases can be confirmed on the radiographs by medical experts and the rest 23 fracture cases are only visible

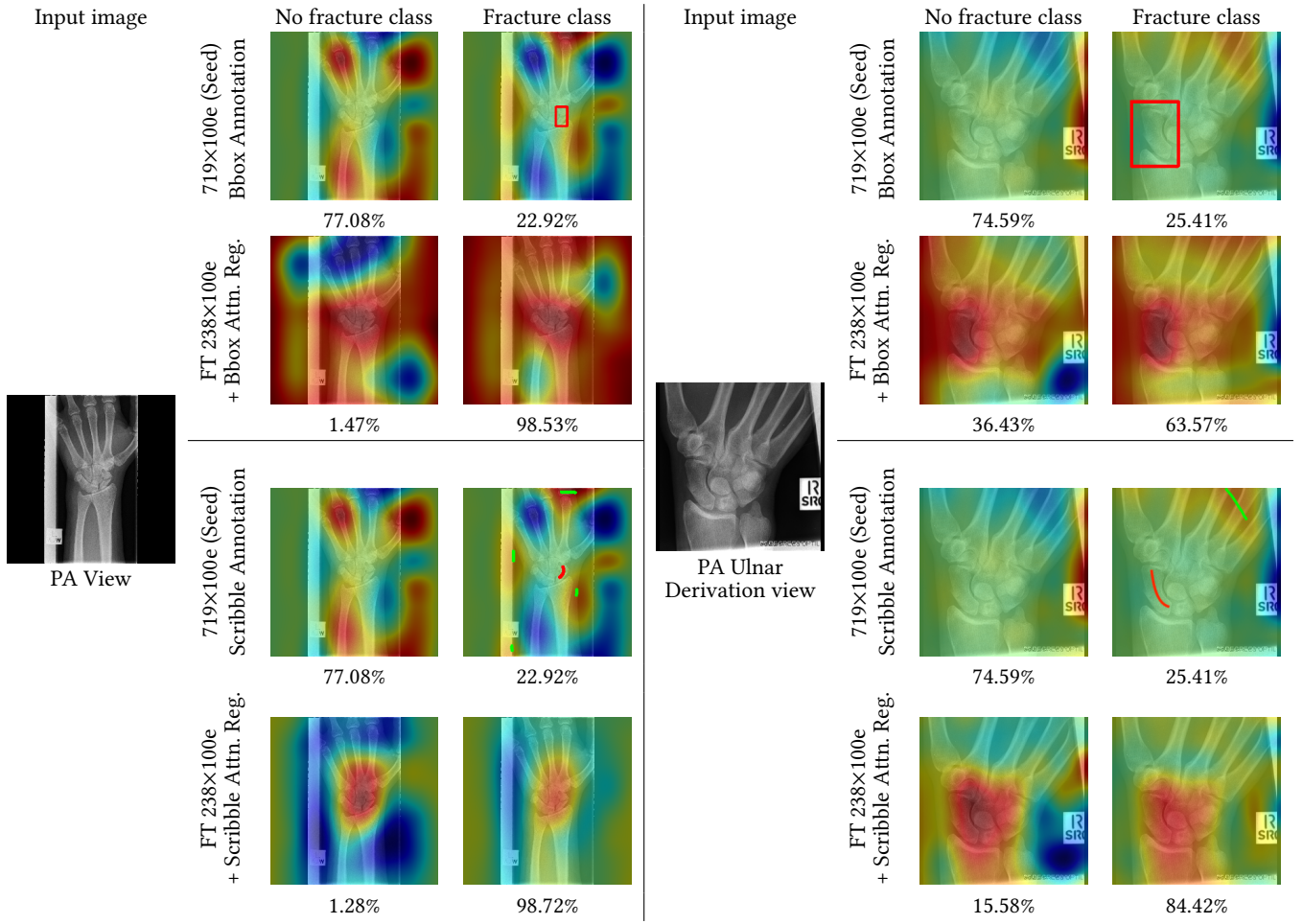


Figure 3: Two examples of the scaphoid fracture validation images showing the change of network attention after fine-tuning with the bounding box style annotation ($\lambda = 0.75$) and the scribble style annotation ($\lambda = 1$). The first row in each block shows the CAM attention maps from the “719×100e” seed model overlaid by the annotation on the ground truth class. The second row shows the CAM after the fine-tuning with the use of the respective annotations. The numbers below each attention image shows the respective classification probabilities.

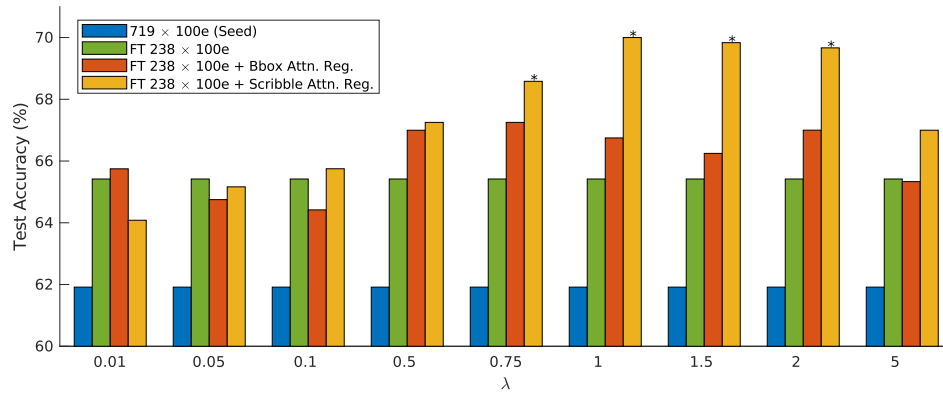


Figure 4: The result comparison on the scaphoid fracture dataset as a function of the λ assigned to the proposed attention regularization. “*” denotes the models which achieve statistically significant improvement over the “FT (fine-tuning) 238×100e” baseline as indicated by t -test with $p < 0.05$.

Table 1: Fundamental differences between the interactive AI (iAI) and HAICS frameworks

	Purpose	Continuousness	Human's options for AI's decision	Start and end points	Performance measurement	Suitable tasks
HAICS	For the human to explicitly provide their expert advice to the AI in order to improve its performance.	Continuous as the AI continuously improves itself based on the continuous feedback loop from the human.	The human can reject a decision from AI and force it to fine-tune itself with their expert feedback if the result is incorrect.	It starts with the AI making a decision based on the trained dataset and ends up with the AI making an improved decision based on the expert feedback from the human (and the process may iterate).	The AI's performance improvement as a result of its sensemaking of the human can be easily measured.	Suitable for supporting pattern recognition tasks where the training data is already available such as image classification.
iAI	For the AI to provide computational support to the human in order to improve their cognitive process.	Not continuous as there is no human feedback loop.	The human has to ingest whatever decision made by the AI and if it is incorrect, they have to manually correct it.	It starts with the human performing actions for a semantic interaction task and ends up with the human ingesting the new information inferred by the AI for this tasks and then moving on to the next task.	It is non-trivial to measure the human's cognitive process improvement.	Suitable for semantic interaction tasks where training data has to be converted from the intercepted ongoing task actions such as document organization.

on the MRI. The diagnosis of fracture or no fracture for each case was confirmed with a corresponding CT or MRI exam.

We use the pre-trained ResNet-50 [12] network model provided by Pytorch. The initial training was carried out with the use of a linear learning rate decay from $1e-3$ to $1e-5$ for 100 epochs using Adam optimizer [14]. The radiographs were first padded to square size and then downsized to 256×256 pixels, the transformation for the input images is a sequence of random cropping (224×224), horizontal flipping, image rotation (i.e., up to ± 30 degrees), and finally resize to 224×224 pixels. Five instances of the ResNet were trained with different random seeds using the 719 training images, resulting a $61.92\% \pm 2.69\%$ test image-level classification accuracy (see the replicated blue “719×100e” bars in Fig. 4).

We first picked a seed model from the “719×100e” models, which has a 60.83% test accuracy. The seed model was uploaded to the browser front-end for network attention inspection. The scribble annotations were made to the 111 incorrectly predicted images in the validation set. We also annotated the same set of images in bounding box style for comparing the two annotation styles. For the bounding box style annotation, the enclosed region inside the red rectangle (see the annotated scaphoid bounding boxes in the first row of Fig. 3) is marked as the positive attention and the outside as the negative attention.

The seed model was further fine-tuned for 100 epochs using the 238 validation images (without the attention annotations) and a linear learning rate decay ($1e-4$ to $1e-5$ for 100 epochs), resulting in a $65.42\% \pm 1.47\%$ test accuracy also computed from five training instances. This result is replicated as the green “FT (finetuning) 238×100e” bars in Fig. 3. With the use of the bounding box (Bbox) style attention annotation, we find that varying the λ value results in two peaks $67.25\% \pm 1.20\%$ and $67.00\% \pm 1.87\%$ at 0.75 and 2 respectively, none of which are statistically significantly improved from the performance of “FT 238×100e”. On the other hand, with the proposed scribble style annotation, we observe in the range

of $\lambda \in [0.75, 2]$, the test accuracy are all statistically significantly improved from the baseline “FT 238×100e” with a single peak of $70.00\% \pm 0.59\%$ when $\lambda = 1$.

4 RELATED WORK

While research on XAI is generally related to this work, the interactive artificial intelligence (iAI) work [34] is most relevant to the HAICS framework. In their work, a blackbox ‘Human Cognition’ in human-AI interaction is identified - *How can the human “explain” him- or herself to the algorithm so that it can respond to his or her internal thought processes, goals, motivations, expert domain knowledge, and intents? How can the machine learn from user interaction?* They refer to the unraveling of this blackbox as explainable cognitive intelligence (XCI).

The work further provides an interactive AI framework where the two blackboxes are connected via two communication channels mediated by a semantic interaction (SI) system for visually organizing relevant textual documents within an open workspace. Human-AI interaction starts with the human manipulating the documents through a sequence of visual interactions such as moving files across the workspace, annotating text, and highlighting words. The AI then detects these actions from which it learns the interests/needs of the user (XCI: AI sensemaking of human). After that, from this learning, the AI discovers other relevant documents and presents them to the human using the same visual metaphors (XAI: human sensemaking of AI).

While this work shares a number of common concepts with the HAICS framework including XAI, XCI, the dual blackboxes, and the dual sensemaking processes, they are fundamentally different as elaborated in Table 1.

It is worth noting that human feedback is also exploited in interactive machine learning (iML) [1, 8, 16, 21]. However, there are subtle differences between HAICS and iML. In HAICS, human feedback is based on AI's explanation to point out AI's incorrect

reasoning and the purpose is for the AI to fine-tune its prediction model with improved reasoning. More specifically, human feedback is provided onto attention maps in the form of scribble annotations for the purpose of improving image classification accuracy. In iML, human feedback is generally based on AI's output to rectify its incorrect result and the purpose is to retrain AI model with new data labelled by the human. Nevertheless, Amershi et al.'s work on the role of humans in interactive machine learning [1] is largely applicable to HAICS as well.

5 CONCLUSION AND FUTURE WORK

In this paper, we present an interactive and continuous sensemaking framework for supporting effective human-AI interaction that has to be established on mutual understanding of each other. While XAI is focused on human sensemaking of AI's reasoning mechanism, the main contribution of this work is on AI sensemaking of human's feedback on its explanations, a critical gap in the current research. Implementation of this framework has been demonstrated by developing a browser-based tool that allows CNN network attention to be visually inspected and annotated by the human in an image classification application. Effectiveness of this framework has been proved through experiments with a real-world scaphoid fracture dataset.

As a proof-of-concept initiative, this work has shown promising results. However, more research is needed to validate the HAICS framework. For the image classification case study, we have proved its effectiveness on AI's performance improvement with one dataset and one iteration of human feedback. More experiments with diverse datasets are required to generalize this finding and particularly to test the effect of continuous feedback. Further work needs to be done to validate the effectiveness of visualizing network attention as heatmaps on human's sensemaking of AI's reasoning mechanisms, particularly with the developed browser-based network visualization tool. In addition, more case studies other than image classification need to be developed to test the general applicability of this framework.

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